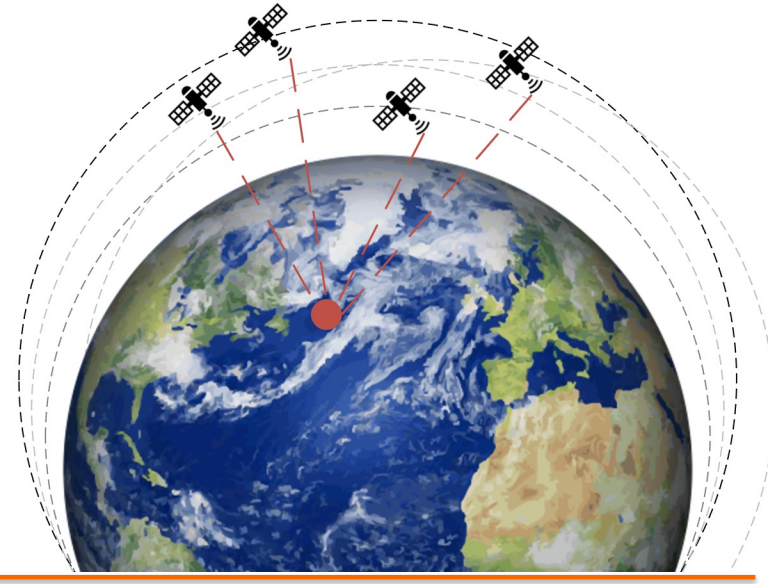
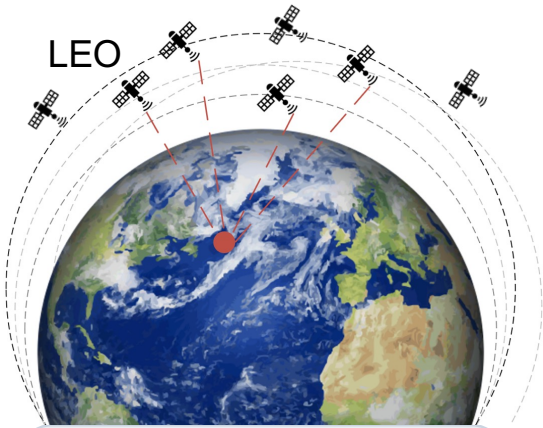
Secure Pseudonymetry with DSSS Watermarking for LEO Satellites



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10/13/2026

IEEE MILCOM
National Capital Region, USA

Problem: A Denser LEO Sky Complicates Source Attribution



Passive RF observation

Opportunistic PNT

Interference monitoring

Today:

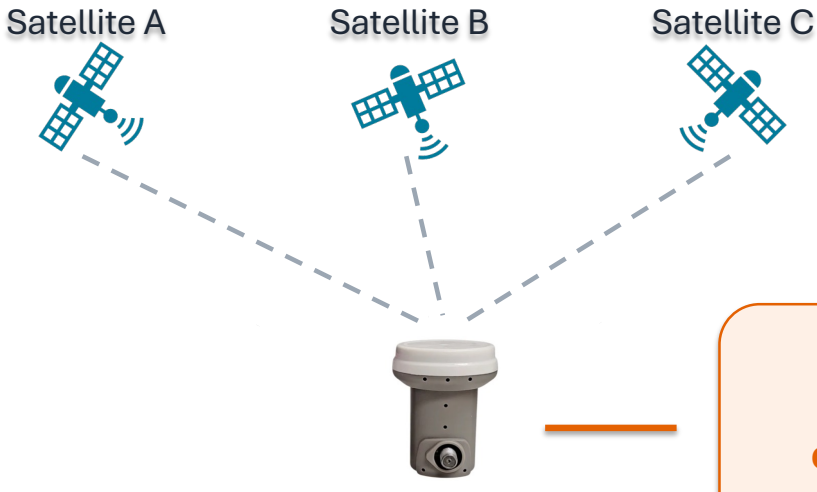
- multiple co-observed satellites
- potential interference

Mega constellations in future:

- even more satellites
- denser interference

1. Signal coexistence
2. Source attribution
3. Authentication

The Missing Piece: Source Attribution



Passive receiver
observes beacon / pilot

Measure navigation observable
Doppler / timing / phase

Associate with closest candidate
$\alpha_{meas} \sim \alpha_i$

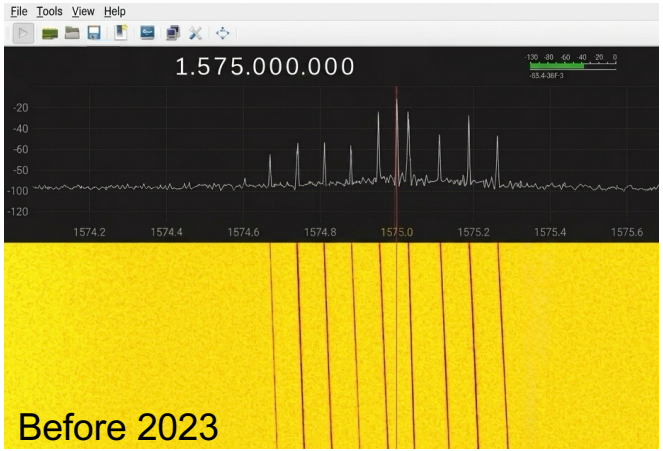
?

Which satellite generated it?

Can the source be trusted?

Detecting an observable ≠ knowing its source

Motivation: Reduced Pilot Power and Coexistence

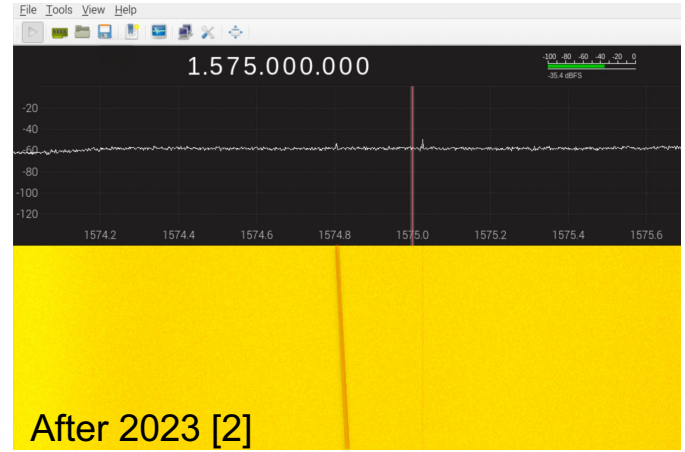


https://forum.amsat-dl.org/index.php?thread/3981-receiving-starlink-satellite-beacons/



single LNB
with 60 dB gain

Pilot tone beacons
reduced power in 2023 [1]



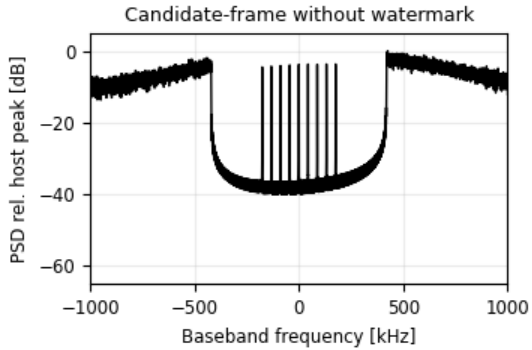
Weaker pilot observability
Post-2023 measurements report ~20–30 dB reduction

Dense multi-satellite setting
Co-observed components can overlap in time and frequency

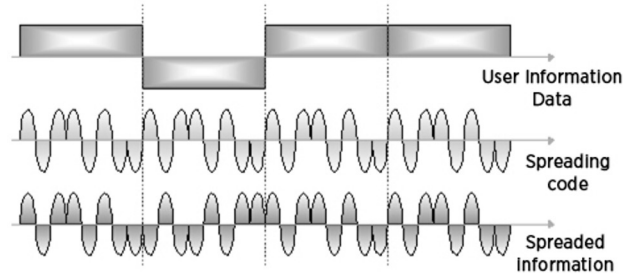
Can we make attribution explicit at the signal layer?

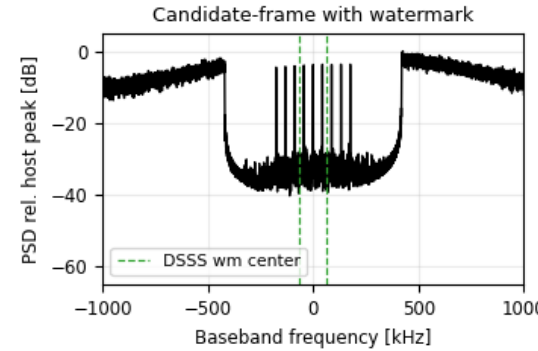
[1] S. Kozhaya, J. Saroufim, and Z. M. Kassas, "Unveiling Starlink for PNT," NAVIGATION: Journal of the Institute of Navigation, vol. 72, no. 1, pp. 1–35, Feb. 2025. doi: 10.33012/navi.685.
[2] N. Camuzcu, T. M. Hoang, and A. Vahid, "Starlink Beacons for Passive LEO-Aided 9D Navigation," IEEE VTC 2026-Fall.

Proposed Idea: Add a Low-Power DSSS Watermark



+





Existing downlink

Primary waveform with pilot tones

DSSS watermark

Low-power transmitter-side overlay

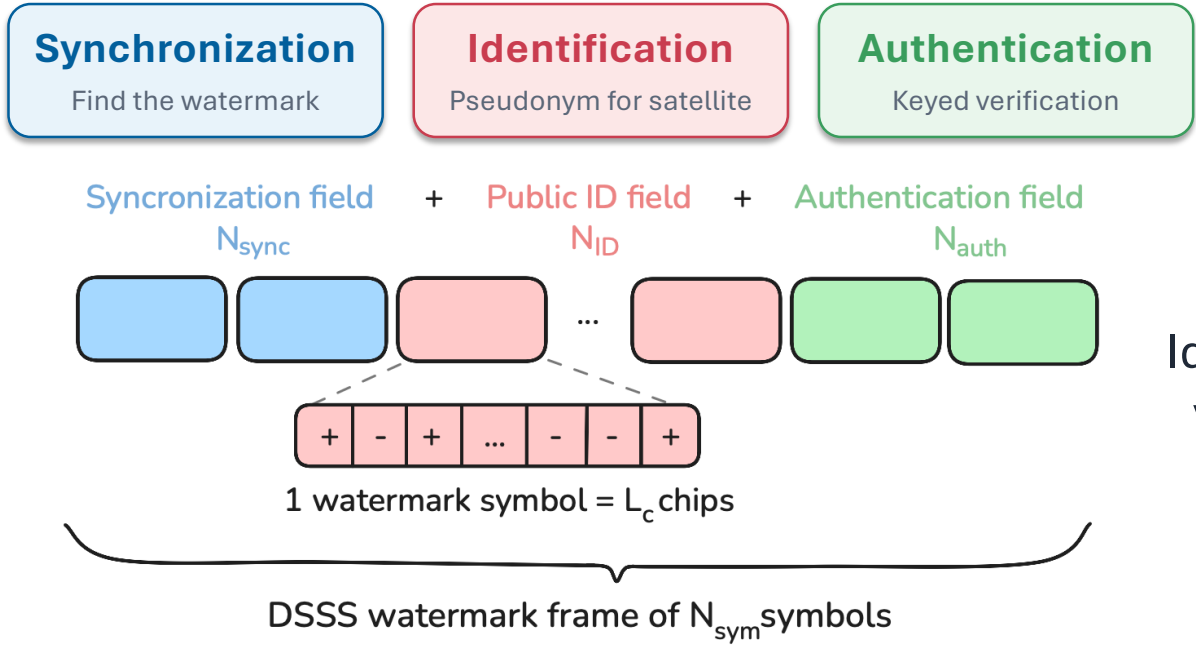
Passive receiver

Correlation recovery
No full payload demodulation

Future transmitter-enabled concept; not part of current Starlink transmissions.

Signal-layer overlay applicable beyond a specific pilot/beacon structure.

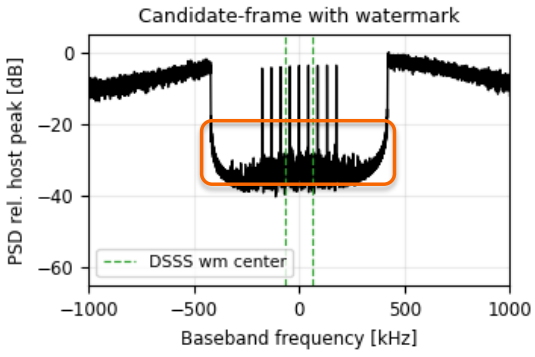
Watermark Frame: Synchronize, Identify, Verify



Identification is public, verification is keyed.

Existing approach
TLE-based association
→
Watermark, no secret key
Identity cross-check
→
Watermark + key
Authenticated attribution

Recovering the Low-Power DSSS Watermark



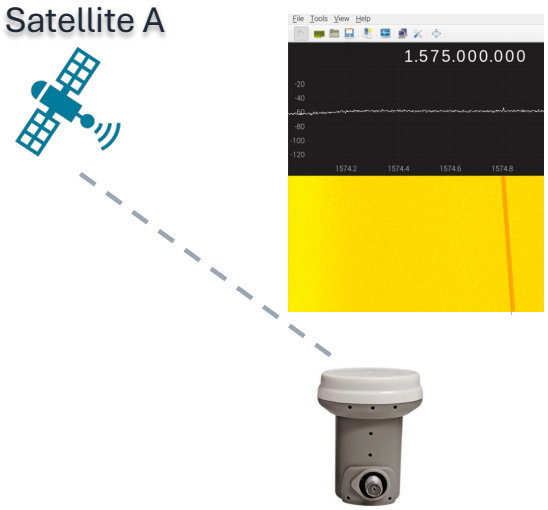
Correlate
Known spreading code
+ delay/CFO search

Despread
Processing gain reveals
the watermark

- Composite downlink = primary signal + DSSS watermark
- Watermark: −35 dB relative to primary → not directly visible

Low transmitted power → Correlation-based recoverability → Backward compatibility

Motion-Aware Passive Receiver



- Motion introduces Doppler:
  Watermark is frequency-shifted and time-varying.

motion CFO

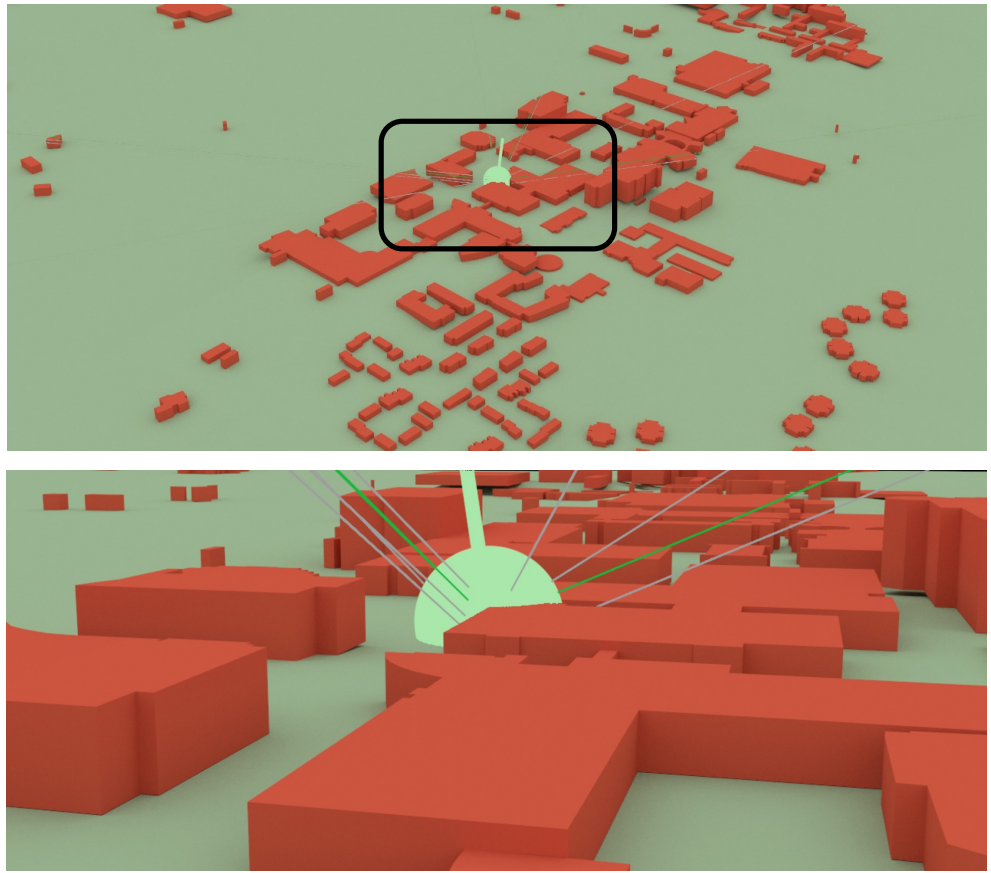
- TLE + beacon assistance:
  To shrink search space before watermark correlation.

1. DSSS acquisition
Timing & frequency search

→

2. Satellite identification + authentication
Public ID + HMAC check

Evaluation: Real Geometry + Local Propagation + Multi-Satellite Mixture



Sionna RT campus scene

8 satellites
same-band coexistence

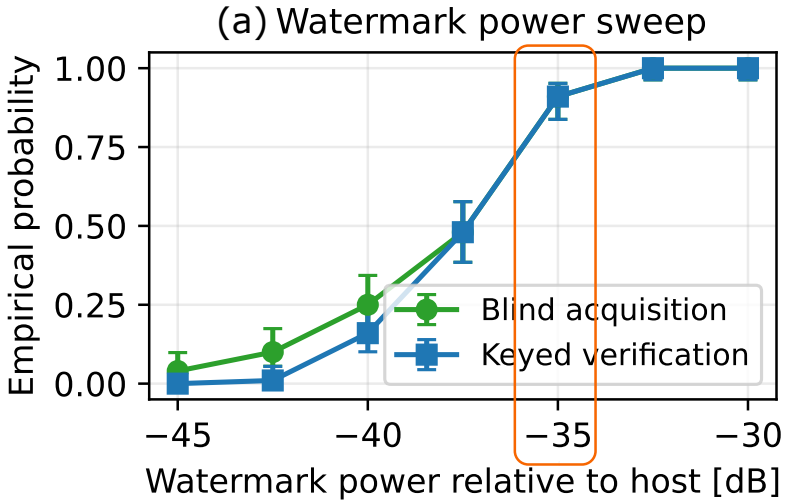
Real Starlink TLEs
geometry + Doppler

400 kchips/s
511-chip code

35 dBi	−25 dB	−35 dB
receiver gain	beacon / host	watermark / host

100 Monte Carlo trials

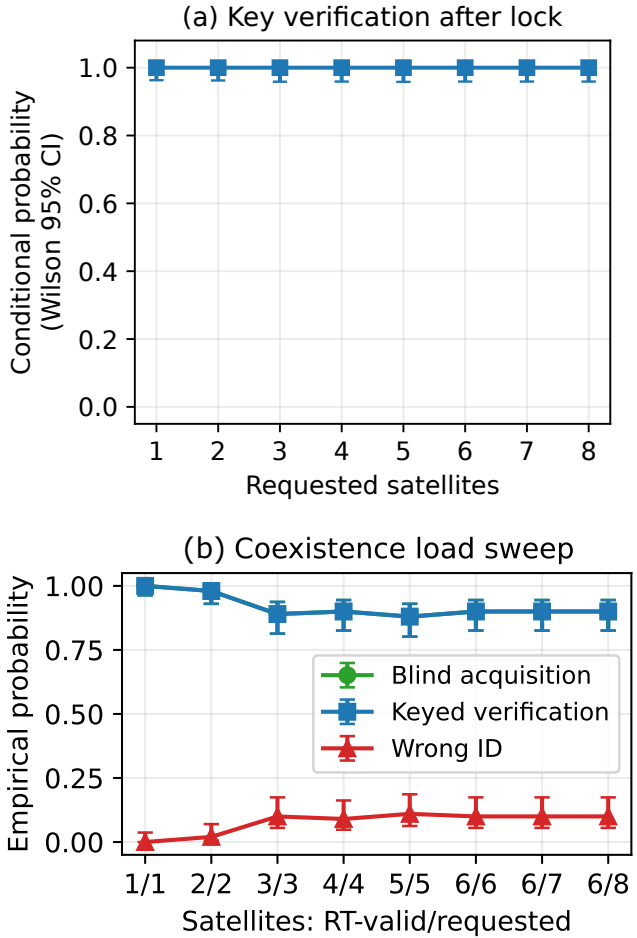
Recovery: DSSS Watermark at −35 dB



- Recovery increases sharply around **−35 dB.**
- Acquisition and verification become unreliable **below −37.5 dB,**

Correlation-based processing gain creates a usable low-power operating region.

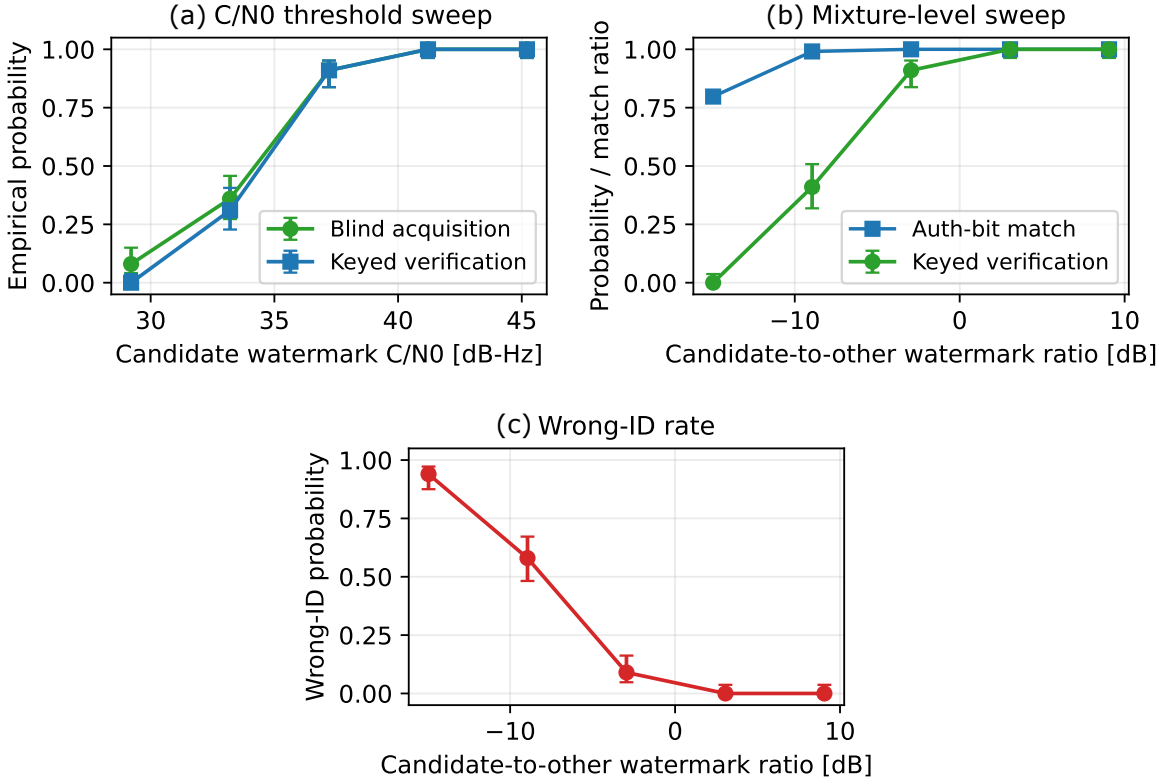
Coexistence: Robust Recovery with 8 Satellites



- **Verification remains around 90%** as the number of co-observed satellites increases.
- Wrong-ID probability remains **around 10%** across the tested load.
- Performance after lock remains **stable.**

Watermark remains distinguishable with up to 8 co-observed satellites.

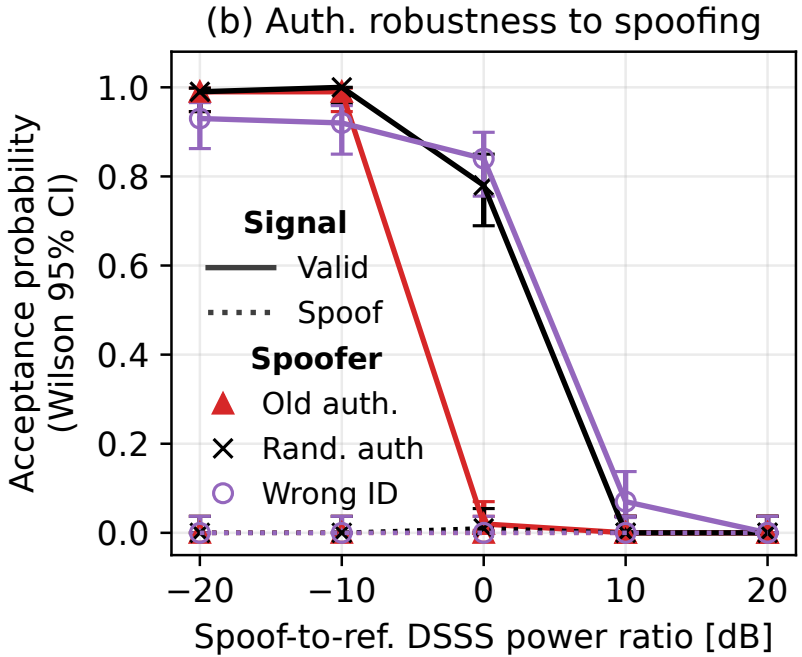
Attribution: Improved by Signal Strength and Separation



- Higher C/N_0 improves acquisition.
- Greater candidate separation improves attribution.
- Wrong-ID probability drops with separation.

Reliable attribution requires sufficient candidate signal strength and separation.

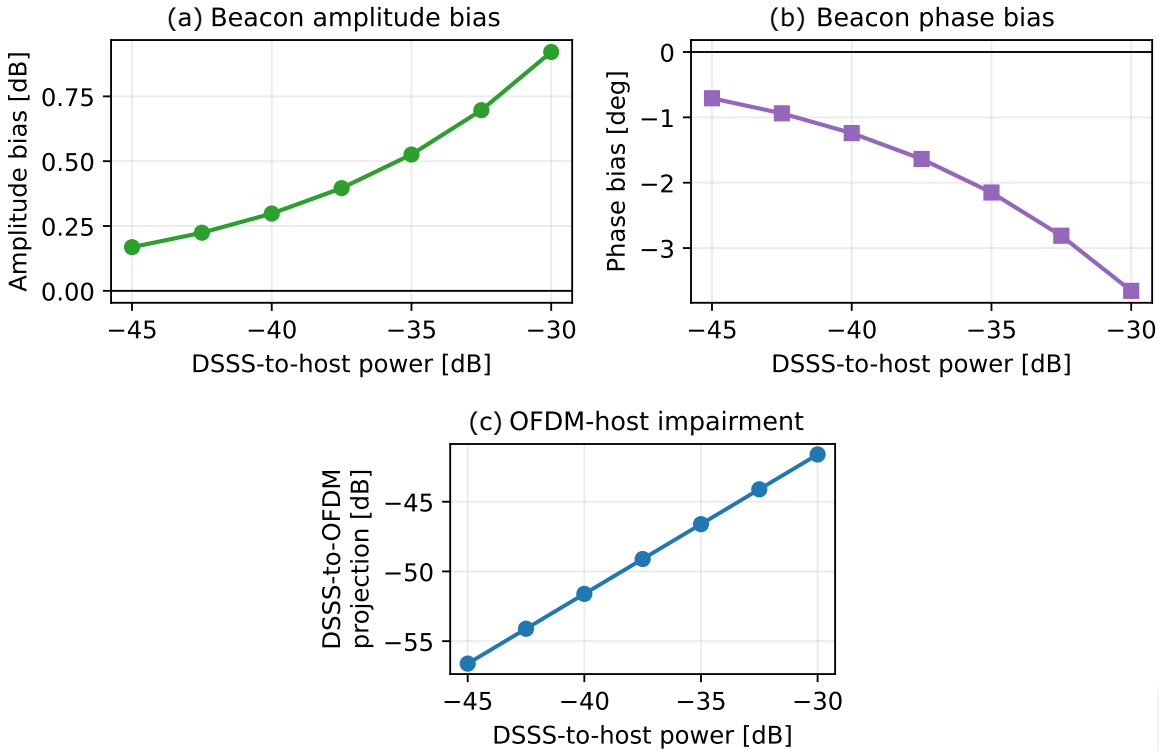
Authentication: Rejecting Invalid Identities



- Random authentication and wrong-ID attempts are rejected.
- Strong interference can still prevent legitimate acquisition.
- Replay requires frame-counter validation.

HMAC verifies source identity, but it does not provide anti-jamming protection.

Compatibility: Minimal Impact on the Primary Downlink



- Beacon impact:
  - ≈ 0.5 dB amplitude bias
  - ≈ 2° phase bias
- Primary-waveform distortion:
  −46.61 dB

Limited perturbation on host with strong watermark recoverability.

Takeaway: From Passive Observable to Verified Source

Without watermark

weak observable

implicit association

no transmitter authentication

→

DSSS pseudonymetry

low-power overlay

public ID

keyed authentication

With transmitter cooperation in future:

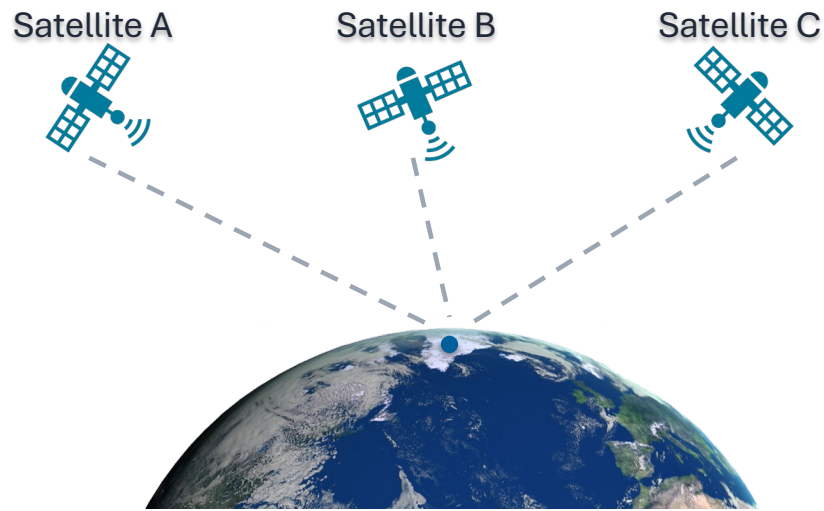
- Low-power recovery
- Explicit attribution
- Authorized verification

Conclusion

A low-power DSSS watermark can provide **satellite identity and keyed authentication** without full payload demodulation **in a multi-satellite environment.**

Future work

Can we go from a candidate identified to interferers identified and mitigated in a shared-spectrum environment?





Thank you!

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References

[1] N. Camuzcu, T. M. Hoang, and A. Vahid, “Starlink Beacons for Passive LEO-Aided 9D Navigation,” IEEE VTC 2026-Fall.

[2] S. Kozhaya, J. Saroufim, and Z. M. Kassas, "Unveiling Starlink for PNT," NAVIGATION: Journal of the Institute of Navigation, vol. 72, no. 1, pp. 1–35, Feb. 2025. doi: 10.33012/navi.685.

[3] T. E. Humphreys, P. A. Iannucci, Z. M. Komodromos, and A. M. Graff, “Signal Structure of the Starlink Ku-Band Downlink,” IEEE Transactions on Aerospace and Electronic Systems, vol. 59, no. 5, pp. 6016–6030, 2023.

[4] M. Neinavaie, J. Khalife, and Z. M. Kassas, “Doppler Stretch Estimation with Application to Tracking Globalstar Satellite Signals,” MILCOM 2021.

[5] W. Qin et al., “Pilots and Other Predictable Elements of the Starlink Ku-Band Downlink,” 2026.

[6] C. J. Hegarty, “GNSS signals — An overview,” IEEE IFCS, 2012.

[7] M. G. Weldegebriel et al., “Watermarking of OFDM for Pseudonymetry: Analysis and Experimental Results,” ICC Workshops, 2024.

[8] H. Krawczyk, M. Bellare, and R. Canetti, “HMAC: Keyed-Hashing for Message Authentication,” RFC 2104, 1997.